Self-Avoiding Walks II: The Hexagonal Lattice, Holomorphic Observables, and the Exact Connective Constant

N-20251127

§1.1 Why the hexagonal lattice is special

For most lattices, the connective constant of self-avoiding walks is not known exactly. The hexagonal lattice is exceptional. Duminil-Copin and Smirnov proved that its connective constant is

$$\mu_{\mathrm{hex}} = \sqrt{2+\sqrt{2}}.$$

Equivalently, the critical step weight is

$$x_c = \frac{1}{\mu_{\mathrm{hex}}} = \frac{1}{\sqrt{2+\sqrt{2}}}.$$

This result is not an enumeration trick in the elementary sense. It comes from a discrete complex-analytic observable. The proof is one of the cleanest examples of the modern two-dimensional philosophy: at the critical point, a lattice model may possess a discrete holomorphic structure that forces exact identities.

Remark 1.1.1 — The number $\sqrt{2+\sqrt{2}}$ should not be memorised as a miracle constant. It is the value for which the parafermionic observable satisfies a local cancellation identity on the hexagonal lattice.

§1.2 When does self-avoidance first change the count?

At very short lengths, the hexagonal lattice makes counting look almost trivial. From a fixed vertex there are three choices for the first step. After that, a walk that merely avoids immediately reversing direction has two possible next steps. A regular hexagon is the shortest cycle in this lattice. Consequently, for $1 \le n \le 5$, a non-backtracking n-step walk cannot revisit any earlier vertex, so

$$\boxed{c_1 = 3, \quad c_2 = 6, \quad c_3 = 12, \quad c_4 = 24, \quad c_5 = 48.}$$

At step six something genuinely new happens. There would be $3 \cdot 2^5 = 96$ non-backtracking paths if we kept ignoring the older part of the walk. But three hexagonal faces meet at the starting vertex, and each can be traversed in two directions to return to that vertex in six steps. Those six paths are forbidden. Hence

$$\boxed{c_6 = 96 - 3 \cdot 2 = 90.}$$

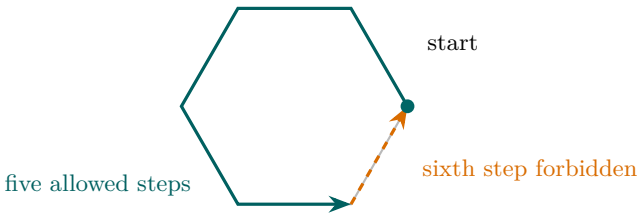


Figure 1.1: A length-five walk around five sides of a hexagonal face visits six distinct vertices; the dashed sixth edge would return to its starting vertex. This is the earliest time that the honeycomb lattice remembers more than the previous step.

This is the difficulty in miniature: a local rule with two immediate choices becomes a history-dependent counting problem as soon as cycles become possible. Finding the exponential growth rate μ requires something stronger than any fixed-length enumeration. The parafermionic observable achieves that by using winding to make entire families of walks cancel.

§1.3 A domain version of the counting problem

Instead of counting walks in the whole plane immediately, one works in a finite domain cut from the hexagonal lattice. Fix a boundary point a. For another boundary point or mid-edge z, consider self-avoiding walks γ starting at a and ending at z, staying inside the domain. Their weighted sum has the form

$$\sum_{\gamma : a \to z} x^{|\gamma|} \times \text{a complex phase depending on the winding of } \gamma.$$

The length weight $x^{|\gamma|}$ remembers the usual self-avoiding-walk generating function. The complex phase is the new ingredient.

The winding of a planar path records how much the direction of the path turns. On the hexagonal lattice the allowed turns are highly constrained, and this makes it possible to tune the phase so that local contributions cancel.

The first picture for this part is especially helpful. On the left, the domain is made of hexagonal-lattice cells and the boundary is described by mid-edges. On the right, the same picture explains winding: the path is not only counted by its length, but also by the total amount it turns.

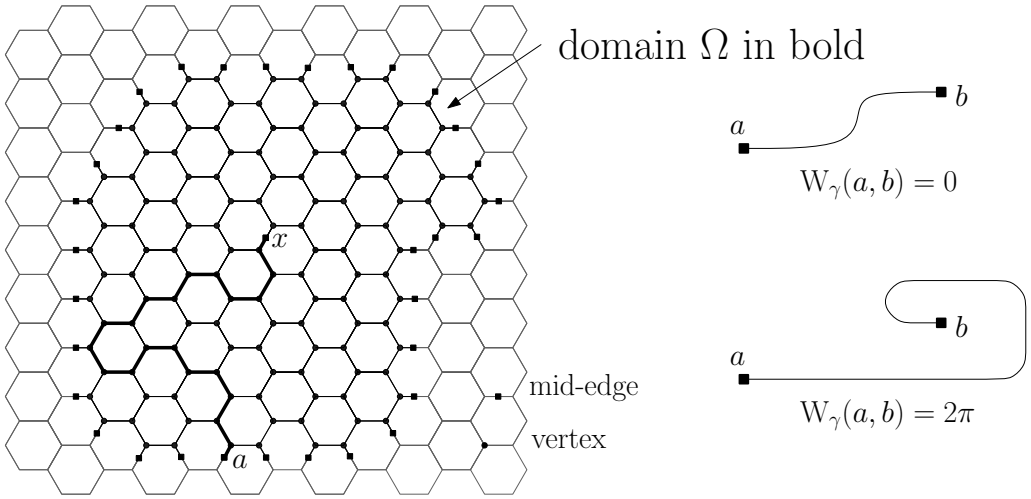


Figure 1.2: a hexagonal-lattice domain and the winding of a self-avoiding walk.

§1.4 The parafermionic observable

The observable can be remembered schematically as

$$F(z) = \sum_{\gamma: a \to z} x^{|\gamma|} e^{-i\sigma W(\gamma)},$$

where $W(\gamma)$ is the total winding and σ is a spin parameter. The exact theorem uses the special choice

$$\sigma = \frac{5}{8}, \qquad x = x_c = \frac{1}{\sqrt{2+\sqrt{2}}}.$$

At this value, the observable satisfies a discrete Cauchy–Riemann type relation around each vertex.

To see where the exact constant comes from, let p, q, r be the three mid-edges touching an interior vertex v, represented as complex points in the plane. The observable satisfies the local relation

$$\boxed{(p-v)F(p) + (q-v)F(q) + (r-v)F(r) = 0.}$$

Duminil-Copin and Smirnov prove it by grouping walks into pairs (that traverse a local loop in opposite directions) and triplets (one walk and its two possible one-step extensions). In a triplet, write

$$j = e^{2\pi i/3}, \qquad \lambda = e^{-5\pi i/24}.$$

After the common length and winding weight is factored out, its contribution to the left-hand side is proportional to

$$1 + xj\overline{\lambda} + x\bar{j}\lambda = 1 + 2x\cos\frac{7\pi}{8} = 1 - 2x\cos\frac{\pi}{8}.$$

This vanishes exactly when

$$x = x_c = \frac{1}{2\cos(\pi/8)} = \frac{1}{\sqrt{2+\sqrt{2}}}.$$

The other paired contributions cancel because of the chosen winding exponent $\sigma = 5/8$. Thus the otherwise unexpected square root is forced by the angles of a hexagon and the cancellation of three possible local paths.[1]

The term *holomorphic* needs a qualification. The displayed identity is a *partial* discrete Cauchy–Riemann relation. It has the useful cancellation property, but by itself does not imply classical holomorphicity of a scaling limit, nor even the existence of that limit. This is exactly why the connective-constant theorem can be proved while convergence of self-avoiding walks to $\mathrm{SLE}_{8/3}$ remains a separate problem.

Remark 1.4.1 — The observable is not merely a generating function. The phase remembers geometry. Without the winding phase, the local cancellation identity would not hold.

§1.5 From local cancellation to a global identity

The local identity is summed over all vertices in a finite domain. Interior terms cancel, leaving only boundary contributions. This is the discrete analogue of using Green's theorem or Cauchy's theorem: local holomorphicity turns an interior statement into a boundary statement.

The cancellation is proved by grouping local contributions. The picture below shows the two types of groupings: pairs of walks that visit all three adjacent mid-edges, and triplets obtained by extending or erasing the last step. At the special values of x and σ, each such group has total contribution zero.

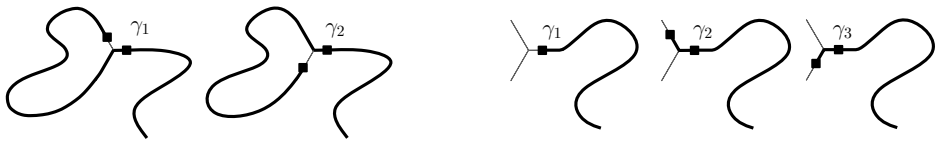


Figure 1.3: pair and triplet cancellations in the proof of the local observable identity.

In a finite trapezoidal strip $S_{T,L}$, partition the endpoints among the left boundary (excluding the start), the right boundary, and the upper and lower boundaries. Call their ordinary, *positive* weighted walk sums $A_{T,L}(x)$, $B_{T,L}(x)$ and $E_{T,L}(x)$, respectively. The domain picture records which walks contribute to which sum.

[1] The derivation follows Duminil-Copin and Smirnov, *The connective constant of the honeycomb lattice equals* $\sqrt{2+\sqrt{2}}$, Sections 2–3.

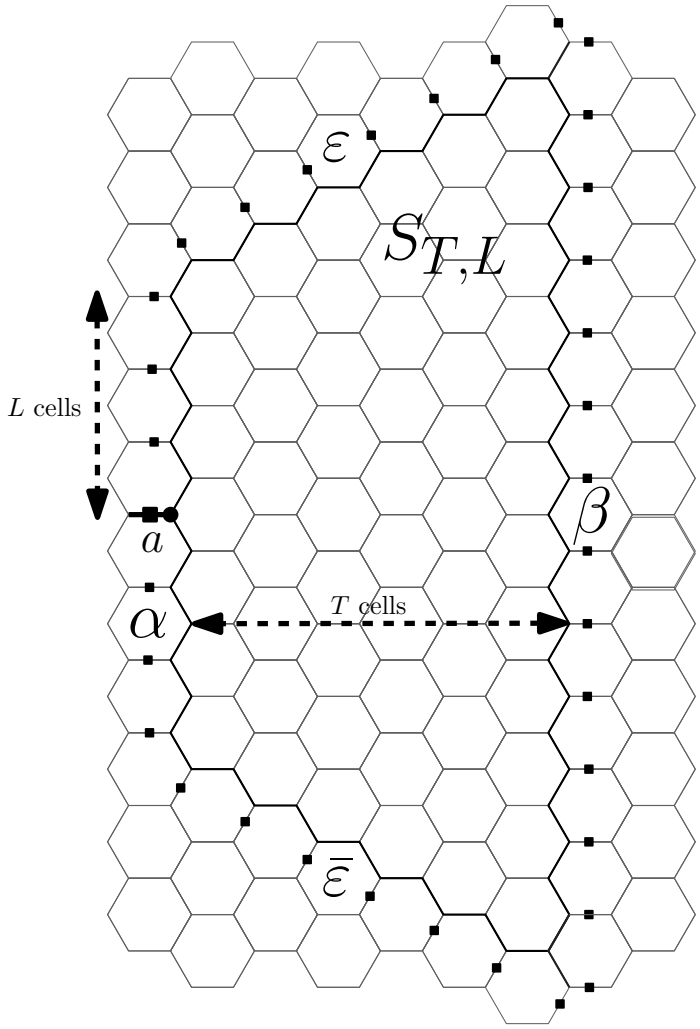


Figure 1.4: the trapezoidal domain $S_{T,L}$ and its boundary pieces.

Summing the local relation over the strip cancels every interior mid-edge. The winding phase on each remaining boundary arc is known, so the complex identity becomes an exact positive real identity:

$$\boxed{1 = \underbrace{\cos\frac{3\pi}{8}}_{c_\alpha} A_{T,L}(x_c) + B_{T,L}(x_c) + \underbrace{\cos\frac{\pi}{4}}_{c_\varepsilon} E_{T,L}(x_c).}$$

All three coefficients are positive. In particular,

$$B_{T,L}(x_c) \le 1, \qquad A_{T,L}(x_c) \le \left(\cos\frac{3\pi}{8}\right)^{-1}.$$

The initial value 1 on the left comes from the empty walk at the starting mid-edge; the trigonometric coefficients are the real parts of its boundary winding weights. The exact coefficients matter, because the same identity will provide both a bound for bridges and a mechanism forcing divergence at the critical point.

As the strip height L increases, the ordinary generating functions for walks reaching a fixed boundary arc have limits, allowing us to work with A_T, B_T and E_T. Their positivity survives the limit.

§1.6 Why divergence gives the lower bound

There is an easy logical trap here. The radius of convergence of the full walk-generating function

$$Z(x) = \sum_{n \geq 0} c_n x^n$$

is $R = 1/\mu_{\text{hex}}$ by the root test. If $Z(x_c)$ diverges, the radius satisfies $R \leq x_c$, and therefore

$$\boxed{\mu_{\text{hex}} \geq x_c^{-1} = \sqrt{2+\sqrt{2}}.}$$

The inequality points this way because taking positive reciprocals reverses order.

The local identity provides a surprisingly concrete route to that divergence. Let A_T, B_T, E_T be the limits of the positive boundary generating functions as the strip height tends to infinity. If $E_T > 0$ for some width T, walks reaching arbitrarily high in that strip already force $Z(x_c) = \infty$: every sufficiently tall boundary contributes at least E_T to disjoint classes of walks.

What if $E_T = 0$ for every width? Then the remaining identity is

$$1 = c_\alpha A_T + B_T.$$

In the original proof, cutting a walk counted by $A_{T+1} - A_T$ at the first visit to the farther edge bounds the difference by two crossing bridges:

$$A_{T+1} - A_T \leq x_c B_{T+1}^2.$$

Subtracting the two boundary identities gives

$$\boxed{c_\alpha x_c B_{T+1}^2 + B_{T+1} \geq B_T.}$$

An elementary induction then implies $B_T \geq c/T$ for some $c > 0$ (one may take $c = \min(B_1, (c_\alpha x_c)^{-1})$). Walks crossing different strip widths form disjoint classes, so

$$Z(x_c) \;\geq\; \sum_{T \geq 1} B_T \;\geq\; c \sum_{T \geq 1} \frac{1}{T} \;=\; \infty.$$

Either way, the generating function diverges at x_c. The harmonic-series bound is a rather beautiful place for a discrete holomorphic identity to end up: all those complex phases have led to a statement about *positive counts*.

§1.7 Why convergence gives the upper bound

For the reverse inequality, we must prove that $Z(x)$ converges at every $x < x_c$. This means $R \geq x_c$, or

$$\boxed{\mu_{\text{hex}} \leq x_c^{-1}.}$$

The critical boundary identity bounds the bridge sum by $B_T(x_c) \leq 1$. A bridge crossing a strip of width T takes at least T steps, so for $0 < x < x_c$,

$$B_T(x) \leq \left(\frac{x}{x_c}\right)^T B_T(x_c) \leq \left(\frac{x}{x_c}\right)^T.$$

Consequently $\sum_{T\geq 1} B_T(x) < \infty$. The bridge decomposition explained in the preceding note represents every walk by two ordered collections of bridges with strictly decreasing widths. This controls the complete partition function:

$$\boxed{Z(x) \leq 2 \prod_{T\geq 1} (1 + B_T(x))^2 < \infty.}$$

The product converges because $\log(1+u) \leq u$ for $u \geq 0$ and the bridge weights have a convergent sum. Thus $R \geq x_c$, as desired.

The two halves of the proof use the boundary identity in different ways: it creates *enough* walks to force divergence at the boundary, while simultaneously placing *strong enough* bounds on crossing bridges to guarantee convergence below it. Together they give

$$\boxed{\mu_{\text{hex}} = \sqrt{2+\sqrt{2}}.}$$

Theorem 1.7.1 (Connective constant of the hexagonal lattice)

For self-avoiding walks on the hexagonal lattice,

$$\lim_{n\to\infty} c_n^{1/n} = \sqrt{2+\sqrt{2}}.$$

§1.8 Relation with SLE

The same observable also points toward the expected scaling limit. The conjectural statement is that two-dimensional self-avoiding walk, after suitable rescaling, converges to $\text{SLE}_{8/3}$. This would imply many predicted critical exponents, such as

$$\nu = \frac{3}{4}, \qquad \gamma = \frac{43}{32}.$$

The exact connective constant result is much more modest than a full scaling-limit theorem, but it belongs to the same conceptual world. The local observable is designed so that at criticality the lattice model begins to look conformally natural.

Remark 1.8.1 — The theorem gives the exact exponential growth constant on the hexagonal lattice. It does not by itself prove the $\text{SLE}_{8/3}$ scaling limit. The former is a rigorous theorem; the latter is the guiding two-dimensional conjectural picture.

§1.9 Loop models and the same mechanism

Self-avoiding walks can be viewed as the $n = 0$ member of the $O(n)$ loop model. In the loop model, configurations consist of nonintersecting loops, and each loop receives a weight n. Formally setting $n = 0$ kills configurations with closed loops and leaves a single open path, which is the self-avoiding walk.

This viewpoint explains why holomorphic observables appear. For special values of the loop weight and the step weight, the model has a parafermionic observable satisfying local relations. The hexagonal self-avoiding walk is the most famous case, but the mechanism belongs to a broader family of two-dimensional lattice models.

The phase diagram situates this in the $O(n)$ phase diagram. For this note, the picture should be read only as orientation: self-avoiding walk corresponds to the $n = 0$

edge of the loop-model story, while the exactly solvable point belongs to a broader two-dimensional critical curve.

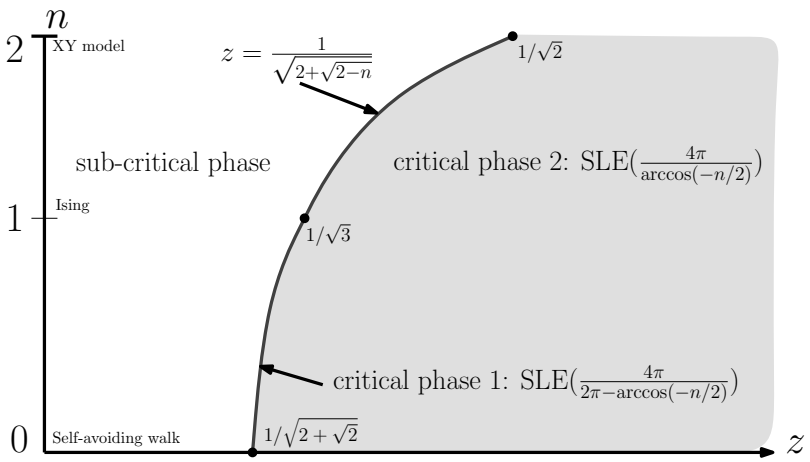


Figure 1.5: phase diagram for $O(n)$ loop models.

§1.10 From example to method

The proof of the hexagonal connective constant has a clear conceptual shape:

choose a winding-weighted observable
$\Downarrow$
tune x and σ so local cancellations hold
$\Downarrow$
sum over a finite domain
$\Downarrow$
obtain positive boundary identities
$\Downarrow$
pass to the infinite lattice and identify μ.

The striking part is that an exact enumerative constant is obtained through a complex-analytic idea rather than through exact enumeration of c_n.